

TERM 3

MATHEMATICS B8 - WEEK 5 (LESSON 1 OF 3)

Week ending: 22 May 2026	Period: 1 of 3	Subject: Mathematics
Duration: 60 mins	Strand: Geometry and Measurement	
Class: B8	Class Size: 42	Sub Strand: Position and Transformation
Topic: Rotation on the coordinate plane and congruence		Lesson in Week: 1 of 3
Content Standard: B8.3.3.1 Perform a single transformation on a 2D shape and describe the image.		Indicator: B8.3.3.1.2-1.3 Draw rotation images and investigate congruent shapes.
Performance Indicator: Learners can draw rotation images of 2D shapes on the coordinate plane and identify properties of congruent shapes.		Lesson: 1 of 3
Core Competencies: Critical Thinking and Problem Solving: Communication and Collaboration: Creativity and Innovation: Digital Literacy		
References: NaCCA Mathematics Curriculum for Basic 8, Mathematics Textbook for JHS Book 2		

Phase/Duration	Learners Activities	Resources
PHASE 1: STARTER 10 mins	Display a picture of the Adinkra symbol 'Dwennimmen' (ram's horns) on the board and ask learners to observe its rotational properties. Ask learners to recall what they learned about rotation in Week 4, including the centre of rotation, angle of rotation, and direction of rotation. Guide learners to identify examples of rotation in their environment, such as the hands of a clock at Accra Mall, rotating ceiling fans, or the wheels of a trotro. Introduce the coordinate plane on the board and review how to plot points using ordered pairs (x, y). Explain that today they will learn to draw rotation images of shapes on the coordinate plane and explore what happens to the shapes after rotation.	Picture of Adinkra symbol Chalkboard/whiteboard Graph book
PHASE 2: NEW LEARNING 40 mins	Draw a coordinate plane on the board and plot a simple triangle ABC with vertices A(2,1), B(4,1), and C(3,3). Demonstrate how to rotate triangle ABC 90° clockwise about the origin (0,0), showing step-by-step how each vertex moves to a new position. Explain the notation for rotation: 90° clockwise can be written as -90° or 270° anticlockwise, using the origin as the centre of rotation. Guide learners to trace the movement of each vertex and plot the image triangle A'B'C', emphasising that the distance from the origin remains constant. Introduce the concept of congruence: explain that the original shape and its rotation image are congruent because they have the same size and shape, only their position and orientation change. Demonstrate how to verify congruence by measuring corresponding sides and angles of the original triangle and its image using a ruler and protractor.	Graph book Ruler Protractor Compass Chalkboard/whiteboard Markers/chalk

	Organise learners into groups of 4 and distribute graph books, rulers, compasses, and protractors to each group. Assign each group to draw a quadrilateral PQRS with vertices P(1,2), Q(3,2), R(3,4), and S(1,4) on their graph books. Instruct groups to rotate quadrilateral PQRS 180° about the origin and draw the image P'Q'R'S'. Ask groups to measure the sides and angles of both the original quadrilateral and its image to verify that they are congruent. Invite 3 groups to present their rotation images on the board, explaining their process and findings about congruence. Provide individual practice: learners draw a rectangle with vertices at (1,1), (4,1), (4,2), and (1,2), then rotate it 90° anticlockwise about the origin. Circulate to provide support, checking that learners correctly apply the rotation transformation and understand that rotation preserves congruence. Summarise the key properties: rotation changes position and orientation but preserves size, shape, and all measurements, making the original and image congruent. Assessment 1. Draw a triangle with vertices at A(1,2), B(3,1), and C(2,4) on a coordinate plane. Rotate it 90° clockwise about the origin and draw the image triangle A'B'C'. What are the coordinates of the vertices of the image? 2. Explain in your own words why a shape and its rotation image are congruent. Give two properties that remain unchanged after rotation. 3. A square has vertices at (2,2), (5,2), (5,5), and (2,5). If you rotate this square 180° about the origin, will the image be congruent to the original square? Justify your answer with at least two reasons.	
PHASE 3: REFLECTION 10 mins	Ask learners to share one new thing they learned about rotation on the coordinate plane and one challenge they faced. Guide learners to reflect on how rotation is different from other transformations they have studied, such as translation and reflection. Pose the question: 'If you rotate a shape twice by 90° clockwise, what single rotation would give the same result?' and discuss responses. Preview the next lesson: explain that in session 2 of 3, they will practise rotating different shapes through various angles and further investigate properties of congruence. Assign homework: Draw any triangle on a coordinate plane and rotate it 90° anticlockwise about the origin, then verify that the original and image are congruent by measuring.	Chalkboard/whiteboard Graph book for homework

TERM 3

MATHEMATICS B8 - WEEK 5 (LESSON 2 OF 3)

Week ending: 22 May 2026	Period: 2 of 3	Subject: Mathematics
Duration: 60 mins	Strand: Geometry and Measurement	
Class: B8	Class Size: 42	Sub Strand: Position and Transformation
Topic: Rotation on the coordinate plane and congruence		Lesson in Week: 2 of 3

Content Standard: B8.3.3.1 Perform a single transformation on a 2D shape and describe the image.	Indicator: B8.3.3.1.2-1.3 Draw rotation images and investigate congruent shapes.	Lesson: 2 of 3
Performance Indicator: Learners can accurately draw rotation images of shapes on the coordinate plane and identify properties of congruent shapes after rotation.	Core Competencies: Critical Thinking and Problem Solving: Communication and Collaboration: Creativity and Innovation: Digital Literacy	
References: NaCCA Mathematics Curriculum B8, Mathematics Textbook for JHS Book 2, Geometry and Measurement Resources		

Phase/Duration	Learners Activities	Resources
PHASE 1: STARTER 10 mins	Display a triangle ABC with vertices A(2,1), B(4,1), C(3,3) on the board and ask learners to recall the rotation performed in session 1. Ask learners to work in pairs to describe what happens to the coordinates of a shape when rotated 90° anticlockwise about the origin. Select 3-4 pairs to share their observations and create a class summary of the coordinate transformation rule for 90° anticlockwise rotation: $(x,y) \rightarrow (-y,x)$. Pose the question: 'If we rotate the same shape 180° or 270°, will the rotated image be congruent to the original? Why?'	Graph board Markers Ruler
PHASE 2: NEW LEARNING 40 mins	Introduce the concept of congruence: shapes that have the same size and shape but may be in different positions or orientations. Demonstrate on graph paper how to rotate a quadrilateral PQRS with vertices P(1,2), Q(3,2), R(3,4), S(1,4) through 180° about the origin, showing step-by-step coordinate transformation using the rule $(x,y) \rightarrow (-x,-y)$. Guide learners to measure corresponding sides and angles of the original shape and its 180° rotation image to verify congruence. Organise learners into groups of 4 and distribute graph books. Each group draws a pentagon on the coordinate plane and rotates it 90° clockwise about the origin using the rule $(x,y) \rightarrow (y,-x)$. Groups investigate whether their original pentagon and rotated image are congruent by measuring all corresponding sides using rulers and all corresponding angles using protractors. Each group records their findings in a table showing original vertices, image vertices, side lengths, and angle measures. Facilitate a class discussion where groups present their findings and conclude that rotation preserves size and shape, making the original and image congruent. Demonstrate a real-life application: rotating the design of a traditional Ghanaian kente pattern on coordinate axes to create symmetrical designs.	Graph books Ruler Protractor Compass Pencils Erasers Coloured pencils

	Learners individually complete a practice exercise: Draw triangle DEF with D(2,3), E(5,3), F(4,6) and rotate it 270° anticlockwise about the origin, then verify congruence by measurement. Introduce the notation for congruence ($\cong$) and how to write congruence statements showing corresponding vertices in correct order. Assessment 1. Triangle ABC has vertices A(1,1), B(4,1), C(2,4). Draw the image of triangle ABC after a rotation of 180° about the origin and state the coordinates of the image vertices. 2. A rectangle has vertices at (2,1), (5,1), (5,3), and (2,3). After rotating it 90° anticlockwise about the origin, is the image congruent to the original rectangle? Explain your answer with reference to at least two properties. 3. Shape KLMN is rotated 270° clockwise about the origin to produce shape K'L'M'N'. If KL = 5cm and angle K = 85°, what are the measurements of K'L' and angle K'? Justify your answer using the properties of rotation.	
PHASE 3: REFLECTION 10 mins	Ask learners to write in their exercise books: 'Three things I learned about rotation and congruence today' and 'One question I still have'. Invite 4-5 learners to share one thing they learned, emphasising the relationship between rotation and congruence. Address common questions or misconceptions identified during the lesson, particularly about coordinate transformation rules for different rotation angles. Preview session 3: learners will apply rotation and congruence concepts to solve complex problems involving multiple transformations and create their own transformation designs. Assign homework: Draw any quadrilateral on the coordinate plane, rotate it 90° clockwise about the origin, and write a paragraph explaining why the two shapes are congruent.	Exercise books Pens

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MATHEMATICS B8 - WEEK 5 (LESSON 3 OF 3)

Week ending: 22 May 2026	Period: 3 of 3	Subject: Mathematics
Duration: 60 mins	Strand: Geometry and Measurement	
Class: B8	Class Size: 42	Sub Strand: Position and Transformation
Topic: Rotation on the coordinate plane and congruence		Lesson in Week: 3 of 3

Content Standard: B8.3.3.1 Perform a single transformation on a 2D shape and describe the image.	Indicator: B8.3.3.1.2-1.3 Draw rotation images and investigate congruent shapes.	Lesson: 3 of 3
Performance Indicator: Learners can apply rotation transformations on the coordinate plane and prove congruence between original shapes and their rotation images through practical investigation and problem-solving.	Core Competencies: Critical Thinking and Problem Solving: Communication and Collaboration: Creativity and Innovation: Digital Literacy	
References: NaCCA Mathematics Curriculum for Basic 8, page 45-48; Ghana Mathematics Series for JHS Book 2, Chapter 12		

Phase/Duration	Learners Activities	Resources
PHASE 1: STARTER 10 mins	Display a coordinate plane on the board with a triangle ABC plotted at coordinates A(2,1), B(4,1), C(3,3) and its 90° clockwise rotation about the origin. Ask learners to work in pairs to identify corresponding vertices between the original triangle and its rotated image. Guide learners to recall from previous sessions: What properties remain unchanged after rotation? (shape, size, angles) Select 3-4 pairs to share their observations about the relationship between the original shape and the rotated image. Introduce today's focus: applying rotation skills to solve real problems and proving that rotated shapes are congruent to their originals.	Graph board Coloured chalk/markers Pre-drawn coordinate grids
PHASE 2: NEW LEARNING 40 mins	Organize learners into groups of 6 (7 groups total) and distribute graph books, rulers, compasses, and protractors to each group. Present a real-world problem: 'A company logo in the shape of a quadrilateral PQRS with vertices P(1,2), Q(3,2), R(3,4), S(1,4) needs to be rotated 180° about the origin for a new design. Draw the original and rotated image, then prove they are congruent.' Guide groups through the problem-solving process: plot the original shape, identify the centre of rotation, apply the 180° rotation rule $(x,y) \rightarrow (-x,-y)$, plot the image P'Q'R'S'. Demonstrate how to prove congruence by measuring corresponding sides and angles using rulers and protractors, recording measurements in a table. Each group completes the problem, then creates their own rotation problem involving a different shape (triangle or pentagon) with rotation angles of 90°, 180°, or 270° about the origin. Groups exchange their created problems with another group and solve each other's problems, verifying congruence through measurements. Facilitate a gallery walk where 2 members from each group remain to explain their work while others rotate to observe different solutions.	Graph books Rulers Compasses Protractors Coloured pencils Problem cards Measurement recording sheets

	Lead a whole-class discussion on patterns observed: Why are rotated images always congruent to originals? What happens to coordinates under different rotation angles? Present a challenge problem: 'A rectangular garden ABCD at coordinates A(2,1), B(5,1), C(5,3), D(2,3) is rotated 90° anticlockwise about point (0,0). Find the new coordinates and calculate the area of both rectangles to verify congruence.' Work through the challenge together, emphasizing that congruent shapes have equal areas, perimeters, and corresponding parts. Assign individual practice: Each learner draws a shape of their choice on the coordinate plane and performs a rotation of their chosen angle, then lists three ways to prove the shapes are congruent.	
	Assessment 1. A triangle with vertices at A(3,2), B(5,2), and C(4,5) is rotated 90° clockwise about the origin. What are the coordinates of the image triangle A'B'C'? Prove that triangle ABC is congruent to triangle A'B'C' by stating three congruence tests. 2. Explain why rotation is called a rigid transformation and how this relates to congruence. Give two real-life examples from Ghanaian contexts where rotation and congruence are important. 3. A quadrilateral WXYZ is rotated 180° about the origin. If the area of WXYZ is 24 cm^2, what is the area of W'X'Y'Z'? Justify your answer using the properties of congruent shapes.	
PHASE 3: REFLECTION 10 mins	Ask learners to complete an exit ticket: 'Write one thing you learned about rotation and congruence this week, one thing you found challenging, and one question you still have.' Invite 4-5 volunteers to share their most interesting discovery about how rotation preserves congruence. Display a summary chart on the board showing: rotation angles and their coordinate rules, properties preserved under rotation (congruence), and methods to prove congruence. Preview next week's topic on data types and collection methods, noting the shift from geometry to handling data. Assign homework: Draw any polygon on a coordinate plane, rotate it 270° anticlockwise about the origin, and write a paragraph explaining how you know the two shapes are congruent using at least three different properties.	Exit ticket slips Summary chart Homework sheets

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